# Alternative Computational Foundation of Chemistry

Claes Johnson

Royal Institute of Technology Stockholm

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**Abstract**

RealQM offers a generalisation of Schrödinger's equation for the Hydrogen atom with $N = 1$ electron to atoms and molecules with $N > 1$ electrons, based on electrons as non-overlapping 3d charge densities subject to Coulomb interaction without self-repulsion and meeting at a common free boundary with continuity and zero normal derivative. RealQM has the form of classical multi-phase continuum mechanics and appears to offer a new tool for ab initio computational quantum chemistry.

## 1 Mathematical Foundation of Chemistry

Chemistry as the science of *molecules* created by *chemical bonding* of *atoms,* seeks its foundation in *atomic physics* as textbook *Standard Quantum Mechanics StdQM* first expressed in mathematical form as *Schödinger's Equation SE* in 1926 [1, 3, 5, 2, 7]. Major efforts over 100 years have been made to explain the physics of chemical bonding by StdQM, but there is still no consensus concerning even the simplest molecule as the *Hydrogen molecule* $H_2$ supposed to be formed by a *covalent bond* between two *Hydrogen atoms* $H$. Nordholm and Bacskay as leading theoretical chemists summarize the situtation in 2020 [4]:

> .... *the concept of a covalent bond ... is still the subject of debate, disagreement, and ignorance with respect to its physical origin.*

The existence of a discipline of Philosophy of Chemistry can be seen as an expression of lack of science.

In this note we present an alternative SE as the basis of *RealQM* as an alternative to StdQM, thus as an alternative foundation of chemistry. The basic concept of RealQM is *electrons* as *non-overlapping 3d charge densities* subject to *Coulomb interaction without self-repulsion* meeting at a common *free boundary* with *continuity and zero normal derivative*. This is a conception compatible with the clear ontology of *classical multi-phase continuum mechanics* with RealQM as natural extension into atomic physics (not StdQM) and is computable in the same sense. RealQM supports a view of the dynamical evolution of a physical system as a form of *time-stepping analog computation* [**?**], which

can be simulated in *digital computation* thus making *computational chemistry* foundational (not in StdQM). RealQM simulation offers force-based explanations of real physics (not StdQM). RealQM can be implemented in a basically 3-line *explicit time-stepping* code capturing the formation of molecules.

We start out returning to the critical moment in 1926 when Schrödinger having formulated his SE for $H$ with one electron, faced the challenge to generalise SE to systems with $N > 1$ electrons. We shall see that the *formalist school* of Bohr-Born-Heisenberg overpowered the *realist school* of Schrödinger-Einstein, and so declared the foundation of chemistry to be StdQM. No alternative was ever formulated and so chemists have for 100 years been left by physicists to make sense of StdQM, while physicists have shifted interest to subatomic physics. RealQM could have been formulated by Schrödinger in 1926, or later or by somebody else, but such an initiative apparently did not surface under the total control of StdQM as expressed by Nobel Laureate Gell Mann at a Nobel Conference in 1976:

- *Niels Bohr brainwashed a whole generation of theorists into thinking that the job (interpreting quantum theory) was done 50 years ago. Niels Bohr brainwashed a whole generation of theorists into thinking that the job (interpreting quantum theory) was done 50 years ago.*

## 2 Outline

We give below a survey of basic feaures of RealQM offering explainations of (1) why $H_2$ forms while $He_2$ and $Au_2$ do not, (2) how $He$ can combine with a proton $H_+$ to form the first molecule $He_+H$, (3) OrthoHelium vs Helium, (4) the periodicity of the *Periodic Table PT* as an *electron packing problem*, (5) (stimulated) atomic emission from charge oscillation between states. RealQM combines computability with clear physical meaning and so opens to *Molecular Dynamics* of large molecules based on physical Coulomb forces.

## 3 Schrödinger 1926: $H$ Atom with $N = 1$

Modern physics of QM was born in January 1926, when the 38 year old Austrian physicist Erwin Schrödinger after an outburst of creativity on vacation in the Swiss Alps together with a friend, rocketed to fame by formulating a mathematical model of the $H$ atom with one electron subject to *Coulomb attraction* from a *positive proton kernel* at $x = 0$ in the form of an *eigenvalue problem* for a *Hamiltonian* as a partial differential operator acting on a real-valued *wave function* $\Psi(x)$ depending on a spatial coordinate $x$ in a 3d *Euclidean space* $\Re^3$, with $\Psi^2(x)$ representing *electron charge density*, named *Schrödinger's Equation SE*.

Schrödinger managed to compute the eigenvalues of SE analytically and found them to agree with Balmer-Rydberg's 1888 formula for the observed spectrum of $H$ and so

could reveal a deep truth about physics expressed in terms of eigenfunctions $\Psi(x)$ determined as stationary points of a *total energy* $E(\Psi)$ given by

$$E(\psi) = E_{kin}(\psi) + E_{pot}(\psi) \equiv \frac{1}{2}\int |\nabla\psi(x)|^2 dx - \int \frac{\psi^2(x)}{|x|}dx \tag{1}$$

with normalisation $\int \psi^2(x)dx = 1$, where $E_{kin}$ is named *kinetic energy* and $E_{pot}$ is *Coulomb potential energy*. Stationarity is attained by $\Psi$ satisfying with $W(x) = -\frac{1}{|x|}$

$$\mathcal{H}\Psi \equiv -\frac{1}{2}\Delta\Psi + W\Psi = E\Psi, \tag{2}$$

with $\mathcal{H}$ a *Hamiltonian* and $E$ *eigenvalue*, which has the form of *classical continuum mechanics* built from a classical Coulomb potential and a stabilising *Laplacian* giving rise to kinectic energy in a form of *localisation energy*. The ground state of $H$ with minimal total energy can be described as a charge density localising optimally around the kernel to decrease potential energy subject to a localisation cost in the form of kinetic energy. In the present context with $\Psi^2(x)$ representing charge density, the kinetic energy represents the new physics of QM, but the terminlogy is misleading, since no motion of electronic charge is present. In any case, SE for $H$ can be solved analytically and so can be viewed to be completely understood from both mathematical and physical point of view. Schrödinger identified the minimal eigenvalue to be $E = E_{kin} + E_{pot} = \frac{1}{2} - 1 = -\frac{1}{2}$ with eigenfunction $\Psi(x) = \frac{1}{\sqrt{\pi}}\exp(-|x|)$ with charge density decaying with $|x|$ from peak at $x = 0$. The balance of $E_{pot}$ and $E_{kin}$ is expressed by the *Virial Theorem* as $E_{kin} = -\frac{1}{2}E_{pot}$.

The complete success with SE for the Hydrogen atom with one electron, asked for a quick generalisation to an atomic system with $N > 1$ electrons. Schrödinger hesitated, but was overpowered by the easy catch suggested by Max Born and Werner Heisenberg in the form of a purely mathematical formal extension by adding a new separate Euclidean space $\Re^3$ for each new electron leading to a wave function $\Psi(x_1, ..., x_N)$ depending on $N$ 3d coordinates $x_1, ...., x_N$, that is a wave function $\Psi(x)$ depending on $x = (x_1, ..., x_N) \in \Re^{3N}$ referred to as *configuration space*, which represents physical space only for $N = 1$.

The formal mathematical extension of SE to $N > 1$ was effortless, and everybody except Schrödinger was happy when Born came up with the idea of giving $\Psi^2(x)$ a probabilistic/statistical meaning covering up lack of direct physical meaning. Schrödinger supported by Einstein protested to letting $\Psi^2(x)$ describe *probabilities of possibilities* and wanted real physics to concern *actualities*. But *statistical physics* had already been introduced in the form of *statistical mechanics* by Boltzmann, and so the road was paved for QM as statistics of electron configurations, and this is still the textbook truth as the *Copenhagen Interpretation CI* of StdQM by Bohr-Born-Heisenberg although sof an *electron* as a *unit charge density without self-repulsion* carrying identity from unique occupancy of a specific region of space, with non-zero electron charge densities meeting with continuity and vanishing normal derivative at the boundary between regions. everal other interpretations have also been suggested. StdQM is viewed as the basic theory of atom physics and so also of chemistry as the science of molecules made of atoms.

But there is a back side to this golden coin: StdQM as account of all possibilities comes with *exponential computational complexity* which has demanded draconic ad hoc dimensional reductions with *Density Functional Theory DFT* the ultimate reduction into one common electron density depending on $x \in \Re^3$ as the only option for larger $N$ based on StdQM, however with unclear physics.

# 4 RealQM for Atoms with $N > 1$

> If we now have to deal with $N$ particles, then $\Psi(x_1, ..., x_N)$ is a function of $N$ variables $x_1$,...,$x_N$ over $N$ 3d spaces $R_1, ..., R_N$. Now first let $R_1$ be identified with the real space $\Re^3$ and integrate over $R_2, ..., R_N$. Second, identify $R_2$ with the real space and integrate over $R_1, R_3, ..., R_N$ and so on. The $N$ individual results are to be added after they have been multiplied by certain constants which characterise the particles. I consider the result to be the electric charge density in real space. (Schrödinger to Lorentz in a letter on June 6 in 1926):
>
> An important historical point is to be noted here. It might have been thought that the correct generalization for N particles is to use N functions of one variable instead of one function of N variables. Such a "wrong turn" did not happen historically, which is, after all, remarkable. (Lieb-Seiringer in Stability of Matter 2005)

To extend SE for $H$ to Helium $He$ with $N = 2$, it is natural from mathematical point of view to split 3d space into two halfspaces meeting at a plane through the kernel and let each electron occupy one half space and meet with continuity of charge density at the diving plane. This is what RealQM does and it shows to give a ground state energy in agreement with observation. We shall see that the split of electrons in He carries through the PT as a "two-valuedness" replacing that of spin.

The same idea for $N > 2$, which is essentially that of the above two quotes, builds on a subdivision of 3d space inoto non-overlapping domains $\Omega_i$ with boundary $\Gamma_i$ for $i = 1, ..., N$, each occupied by a non-zero electron charge density, which meet with continuity on the common total boundary $\Gamma$ as the union of all $\Gamma_i \bigcap \Gamma_j$ with $i, j = 1, ..., N$. We let $H^1(\Omega)$ denote the set of real-valued functions which are square integrable together with first partial derivatives over a domain $\Omega$. RealQM thus builds on a natural mathematical conception of electrons as a charge densities occupying non-overlapping regions in space and meeting with continuity, which can also be given a clear physical meaning in terms of classical continuum physics.

RealQM for an atom takes the following form restricting here to *ground state* with *minimal energy*: Find the function $\psi(x) = \sum_{i=1}^N \psi_i(x)$ in $H^1(\Re^3)$ with $\psi_i \in H^1(\Omega_i)$ and $\int \psi_i^2(x)dx = 1$, which minimizes the *total energy*

$$TE(\psi) \equiv K(\psi) + PK(\psi) + PE(\psi)$$

as the sum of *kinetic energy*:

$$K(\psi) = \frac{1}{2}\int_{\Re^3} |\nabla\psi(x)|^2\,dx,$$

*attractive kernel potential energy*:

$$PK(\psi) = \int_{\Re^3} W(x)\psi^2(x)\,dx,$$

and *repulsive electronic potential energy*:

$$PE(\psi) = \sum_i \int_{\Omega_i} \sum_{j\neq i} V_j(x)\psi_i^2(x)\,dx,$$

where $W(x) = -\frac{Z}{|x|}$ is the potential generated by a pointlike positive kernel of charge $Z$ and $V_j(x)$ is the potential (modulo factor 2) generated by electron charge density $j$ defined by

$$V_j(x) = \int_{\Omega_j} \frac{\psi_j^2(y)}{2|x-y|}dy \quad \text{for} \quad x \in \Re^3.$$

We understand that the kinetic energy appears as a measure of charge density variation in space and does not express any motion of charge in space, but rather as an analog to elastic energy connected to strain. In particular, the ground state charge density has no time dependence, which means that the ground state does not radiate energy and so can persist over time. On the other hand, a radiating dipole in resonance with external forcing can be created as oscillation bewteeen ground and excited state. The model contains *no electronic self-repulsion* since $j \neq i$ in the summation to $PE(\psi)$. This is an essential difference compared to StdQM/DFT, where effects of overlapping wave functions including self-repulsion need attention.

Variation of $TE(\psi)$ over $\psi = \sum_i \psi_i \in H^1(\Re^3)$ with $\psi_i \in H^1(\Omega_i)$, leads to the following characterization of a minimiser $\Psi = \sum_i \Psi_i \in H^1(\Re^3)$ with $\Psi_i \in H^1(\Omega_i)$ and $\int \Psi_i^2(x)dx = 1$, satisfying for $i = 1, ..., N$:

$$\mathcal{H}_i\Psi_i \equiv -(\frac{1}{2}\Delta + W + 2\sum_{j\neq i} V_j)\Psi_i = E_i\Psi_i \text{ in } \Omega_i, \tag{3}$$

$$\frac{\partial\Psi_i}{\partial n} = 0 \text{ on } \Gamma_i, \tag{4}$$

where $E_i$ is the Lagrange multiplier for electron charge $\int \Psi_i^2\,dx = 1$, and free variation of $\psi_i$ over $\Omega_i$ imposes the homogeneous Neumann boundary condition (4) with $n$ normal to the boundary. The boundary $\Gamma$ appears as a *Bernoulli free boundary* combining a homogeneous Neumann condition and a Dirichlet/continuity condition. RealQM can be seen as a form of DFT [5] based on one-electron non-overlapping charge densities forming a total charge density. RealQM also connects to Bader's AIM [7] but differs by being based on charge densities rather than basins of kernel attraction.

## 5 3-Line Code

The ground state energy $TE(\Psi)$ of RealQM is computed by a gradient method including the following 3 steps:

1. Each electron seeks to minimize its contribution to total energy subject to Coulomb interaction with the other electrons and the kernel as expressed by (3) and (4).
2. The potentials $V_j$ are computed as solutions the Poisson equation $-\Delta V_j = 2\pi\psi_j^2$ in $\Re^3$.
3. The free boundary is updated in terms of *level set functions* $\chi_i(x)$ serving as characteristics functions of each $\Omega_i$ as solutions to level set equations with coefficients $\beta_i$ depending on jumps of wave functions across common boundaries moving the free boundary in the direction of lower charge density and so towards continuity.

A Code Repository [21] contains benchmark problems computed by realising 1-3 in the form of explicit time-stepping of the following equations on a fixed 1 million mesh points spatial grid as a 3-line code: For $i = 1, ..., N$,

$$\dot{\Psi}_i + \mathcal{H}_i\Psi_i = 0 \quad \text{in } \Omega_i,\ \frac{\partial\Psi_i}{\partial n} = 0 \quad \text{on} \quad \Gamma_i \tag{5}$$

$$\dot{V}_i - \Delta V_i = 2\pi\Psi_i^2 \quad \text{in } \Re^3, \tag{6}$$

$$\dot{\chi}_i + \beta_i|\nabla\chi_i| = 0 \quad \text{in } \Re^3, \tag{7}$$

where the dot signifies derivation with respect to time, and in each time step $\Psi_i$ is normalised to unit charge. Computations typically are initialized with electrons well separated allowing the free boundary to develop dynamically with an overlap of level set functions of a couple of mesh points, and so the creation of a molecule can be followed. The codes in Code Repository can directly be tested and modified on the web.

RealQM directly extends to molecules by adding more kernels. Formation of a molecule can be computed by evaluating forces on kernels from electron charge potentials and moving kernels until equilibrium. More generally RealQM opens to *Molecular Dynamics MD* based on physical forces from electron potentials, instead of the energy based pseudo-forces using StdQM. RealQM thus may offer a new tool for ab initio simulation of protein folding, where the only parameter is the mesh resolution free of any choice of basis functions.

## 6 Covalent Bonding

Chemistry can be seen as the physics of chemical bonding of atoms into molecules with lower energy than separate atoms, described as either *covalent* with "sharing of electrons" or *ionic* with "transfer of electrons". The unclear physicality of StdQM offers an unclear explanation of covalent bonding as (A1) localisation of electrons beween kernels decreasing potential energy and (B1) delocalisation of electrons over the whole molecule

to avoid increase of kinetic energy from localisation. But localisation and delocalisation appears to be contradictory from physical point of view and so (A1)+(B1) may not describe real physics and not even consensus.

RealQM explains covalent bonding as (A2) localisation of electrons beween kernels decreasing potential energy and (B2) free boundary allowing electrons to meet with non-zero charge and so without increase of kinetic energy from localisation. There is no contradiction between (A2) and (B2) and so (A2)+(B2) appears to represent physics better than (A1)+(B1). We first show RealQM applied to a 2d model molecule $X_2$ formed by a covalent bond between two Hydrogen like atoms $X$ with one electron surrounding a kernel. We compute the total energy at different distances between the kernels to discover that the total energy decreases with decreasing distance until the kernel repulsion takes over with an increase of total energy. We analyse the contributions to total energy from kernel-electron, electron-electron and kernel-kernel potential energy and electronic kinetic energy to understand the dip to a minimum of total energy at a certain kernel distance forming the covalent bond.

We see in the screenshots of computations below in 2d wave functions (red) and 1d cross-cuts of wave functions (red), kernel potentials (blue) and electron potentials (light-blue). We identify in particular the shaded area between the blue and light-blue curves which combine with the red to give a net decrease of potential energy with negative kernel-electron energy dominating positive electron-electron energy. We see that the wave functions meet with non-zero value allowing kinetic energy to stay constant. In the plots, potential energy refers to the combined kernel-electron and electron-electron energy. It is natural to name the chemical bond formed by the meeting of electron wave functions at a free boundary between the kernels as a *covalent bond*, but then with new meaning compared to StdQM.

Figure 1

Figure 2

Figure 3

We now turn to the basic example of formation of a $H_2$ molecule which can be followed by running the RealQM 3d code [25] to discover a stationary state after explicit time-stepping as a $H_2$ molecule at a kernel distance of 1.4 atomic units. We can follow charge concentration forming between the kernels with wave functions meeting at a midway plane orthogonal to the line between kernels, in a process without increase of kinetic energy just like in the model case above. RealQM thus delivers a covalent bond forming $H_2$.

The energy is reported here as the same bracket as for $He$ below, and for the same reason. The two ways of imposing the exclusion at $N = 2$ give $-1.1336$ with the electrons

sharing one density and $-1.2750$ with them on disjoint half-spaces, against the reference $-1.174476$: a shared density under-binds by 3.5%, disjoint domains over-bind by 8.6%, and the reference lies inside the span of the model's own two treatments. The bracket is wider than helium's, and no dissociation energy is quoted, because a dissociation energy would assert a position inside it. (An earlier version of this passage quoted $-1.17$ Hartree, and a dissociation energy of 0.17 Hartree, as agreeing with the list values. That is withdrawn. The value came from a run on a much coarser mesh, $h = 0.13$–$0.20$, on which two large errors cancel: the pair repulsion is 7% low, which over-binds, while the coarse grid together with the explicit $r \to \sqrt{r^2 + h^2}$ kernel softening under-binds. Refinement removes the cancellation, and the axisymmetric reduction — which is exact for two domains on an axis and carries no angular discretisation error — converges the disjoint-domain construction to $-1.276$. So $-1.17$ is a value the relaxation passes *through* on its way to the construction's own minimum, exactly as $-2.91$ is for helium.)

With the same code we can follow the formation of Helium $He$ letting the kernels coincide while neglecting kernel-kernel repulsion, which gives an energy of $-2.86$ with the two electrons sharing one density and $-3.03$ with them on disjoint half-spaces, bracketing the reference $-2.9037$ from either side. The two ways of imposing the exclusion at $N = 2$ fail in opposite directions — a shared density under-binds by 1.5%, disjoint domains over-bind by 4.2% — so the reference value lies inside the span of the model's own two treatments. (An earlier version of this passage quoted $-2.91$, within 0.24%, and said the agreement could be made exact by giving the kernel a small radius. Both are withdrawn: $-2.91$ is a value the relaxation passes *through* on its way to the construction's own minimum, and the kernel softening that reproduces $-2.9037$ scales as the mesh spacing, so it vanishes in the continuum and is a numerical constant rather than a physical kernel size.)

Having understood how two $H$ atoms can form a $H_2$ molecule by a covalent bond as well as $He$, we now ask if $He$ can similarly form a $He_2$ molecule. We run the code [27] to discover that this is not so, because the charge concentration between kernels is not taking place, because the two planes through the kernels separating the two electrons of each He atom, are pushed away from the center to maintain continuity of charge density at the separating planes. RealQM delivers the following total energies E for two He atoms at varying kernel distance D= 4, 3.2, 2.4, 1,6, E= 0.00, -0.04, 0.03, 0.20, showing very little bonding energy and so gives evidence that $He$ does not form a $He_2$ molecule.

Figure 4: Non-existence of He2 molecule by non-accumulation of charge density between kernels by outward shift of planes P separating the electrons for each He atom to maintain continuity at P.

## 7 The First Molecule $He_+H$

Standard Cosmology says that Big Bang created free electrons and free protons forming neutrons together forming the Helium kernel from 2 protons and 2 neutrons, which then

combined with electrons into the first atom as $He$, thus forming before combination into $H$, because of stronger kernel attraction. The first molecule was then formed by compressing a proton $H+$ and $He$ to form $He_+H$ by transfer of an electron from $He$ to $H_+$ in a weakly exothermic reaction, which can then dissociate into $He_+$ and $H$, thus altogether a process of forming $H$ from $He$ in the early universe. [1, 6].

Figure 5

Figure 6

We now let RealQM model this process using this code [28] to find $He_+H$. We see here the characteristic feature of RealQM for Helium with its two electrons occupying two separate non-overlapping half lobes meeting at a plane together with the +2 kernel in the middle, and we also see a proton inserted on the left. The energy of the system is the ground state energy E = -2.903 of $He$ leaving the energy required for the proton insertion to the simulation. We watch the dynamics when the left electron is attracted to the proton and the right electron gradually takes over the Helium kernel ino a final state with energy E = -2.490 depicted in Fig. (6).Recall the shooting off an electron from He requires 0.903 Hartree to create $He_+$ which is a lot, while the above process with instead a proton capturing an electron involves only 0.413 Hartree from -2.903 to -2.490 with the rest borrowed from H.

## 8 Helium and OrthoHelium

A Helium atom in ground state has 2 electrons surrounding a +2 kernel. In StdQM the two electrons are overlapping with different spin. In RealQM electrons do not overlap (spin is not an issue) and so are restricted to two half spaces meeting at a plane through the kernel. The formation of the He ground state can be followed running the code [26] delivering a total energy of $-3.03$ Hartree with the electrons on disjoint half-spaces, against the reference $-2.9037$; the shared-density alternative gives $-2.86$, so the reference lies between the two. We can see He with its two halfspaces being formed by eliminating kernel repulsion in H2 letting the kernel distance go to zero [14].

Figure 7: Orthohelium as inner(red) and outer(green) shell electrons.

The first line in the spectrum of He corresponds to an energy of 0.728 Hartree as the difference between a 1st excited state with energy $-2.175$ Hartree and the ground state with energy $-2.903$. The excited is referred to as *Orthohelium* to be compared with a state of *Parahelium* with a larger energy of $-2.145$ Hartree. Orthohelium shows a stronger line than Parahelium.

Figure 8: Orthohelium as excied half-space electrons.

We now let RealQM compute an state of He with one of the electrons excited to an outer shell with the remaining electron filling an an entire inner shell around the kernel. Running the code [10] we find that RealQM gives an energy of $-2.175 \pm 0.005$ in agreement with Orthohelium., see Fig 7.

We compare using the code [11] excitation keeping the half-space geometry of the ground state to find again an energy of $-2.175 \pm 0.005$ Hartree, see Fig. 8. RealQM thus delivers Orthohelium as the 1st excitation of He in two geometrically different settings, in agreement with a strong line.

StdQM presents Orthohelium as a complicated triplet state with the electrons having same spin, and Parahelium as a simple singlet state with the two electrons having opposite spin.

# 9 Periodic Table from Electron Packing

RealQM presents the ground state of an Atom as the result of an *electron packing* energy minimization problem, where each electron seeks to minimize its contributtion to the total energy under Coulomb attraction/repulsion from the atom nucleus and the other electrons, while the supports of the electrons meet at a common *free boundary* determined so as to maintain *continuity of electron charge density*. In this packing problem electrons take on a certain width/volume in 3d space determined by proximity to the kernel while competing for space with other electrons. RealQM can thus be seen as a "ballon packing" problem with electron kinetic energy serving as a measure of "compression".

RealQM packs electrons into an expanding sequence of on-overlapping spherical shells $S_n$ centered at the kernel with $n = 1, 2, 3...$ with each $S_n$ first subdivided into two half-spherical shells each of which is divided into two angular directions into $n \times n$ electron domains thus with a total of $2n^2$ electrons in each full shell $S_n$. This gives the familiar sequence 2, 8, 18, 32,... which is repeated in the periods of the Periodic Table 2, 8, 8, 18, 18, 32, 32.

In particular, RealQM suggests the following shell structures of filled shells with 8-shells radially divided into two 4-shells:

- (2)+(4+4)
- (2)+(4+4)+(4+4)
- (2)+(4+4)+(8)+(2)
- (2)+(4+4)+(18)+(2)
- (2)+(4+4)+(18)+(8)
- (2)+(4+4)+(18)+(18)+(4+4)

- (2)+(4+4)+(18)+(32)+(18)+(4+4)

Below we show computed ground state energies with code [22] (here reduced to 3 shells), where the electrons in each subshell are homogenised to spherical symmetry with the electron subshell structure indicated and table values in parenthesis. The shell configration is partly up to choice with best choice in principle having smallest energy, but the choice also has to be accessible in a dynamic build of the atom by succesively adding electrons. This opens a rich world to explore by RealQM.

- Lithium (2)+1: -7.55 (-7.48) 1st ionisation: (0.2)
- Beryllium (2)+(2): -15.14 (-14.57) 1st ionisation: 0.5 (0.35)
- Boron (2)+(2+1): -25.3 (-24.53) 1st ionisation: 0.2 (0.3)
- Carbon (2)+(2+2): -38.2 (-37.7) 1st ionisation 0.5 (0.4)
- Nitrogen (2)+(3+2): -55.3 (-54.4) 1st ionisation 0.5 (0.5)
- Oxygen (2)+(3+3): -75.5 (-74.8) 1st ionisation 0.5 (0.5)
- Fluorine (2)+(3+4): -99.9 (-99.5) 1st ionisation 0.5 (0.65)
- Neon (2)+(4+4): -132.4 (-128.5 ) 1st ionisation 0.6 (0.8)
- Sodium (2)+(4+4)+(1): -165 (-162)
- Magnesium (2)+(4+4)+(2): -202 (-200)
- Aluminium (2)+(4+4)+(2+1): -244 (-243)
- Silicon (2)+(4+4)+(2+2): -291 (-290)
- Phosphorus (2)+(4+4)+(3+2): -340 (-340)
- Sulphur (2)+(4+4)+(4+2): -397 (-399)
- Chlorine (2)+(4+4)+(3+4): -457 (-461)
- Argon: (2)+(4+4)+(4+4): -523 (-526)
- Calcium: (2)+(4+4)+(8)+(2): -670 (-680)
- Titanium: (2)+(4+4)+(10)+(2): -848 (-853)
- Chromium: (2)+(4+4)+(12)+(2): -1039 (-1050)
- Iron: (2)+(4+4)+(14)+(2): -1260 (-1272)
- Nickel: (2)+(4+4)+(16)+(2): -1516 (-1520)
- Zinc: (2)+(4+4)+(18)+(2): -1773 (-1795)

- Germanium: (2)+(4+4)+(18)+(2+2): -2089 (-2097)
- Selenium: (2)+(4+4)+(18)+(4+2):- 2416 (-2428)
- Krypton: (2)+(4+4)+(18)+(4+4): -2766 (-2788)
- Xenon: (2)+(4+4)+(18)+(18)+(4+4): -7355 (-7438)
- Radon: (2)+(4+4)+(18)+(32)+(18)+(4+4): -22800 (-23560)

Figure 9: Basic shell structure

We see good agreement even with the relatively crude approximation of homogenisation into spherical symmetry used in the computations. In Figs. (10) and (11) we see interface and output for Oxygen in 2+3+3 configuration with 2+4+2 and 2+2+4 of similar energy.

To see the effect of the subshell structure we compare Neon: (2)+(4+4) with Neon: (2)+(8) without the (4+4) subshell structure, which has a ground state energy of -153, which is much smaller than the observed -128.5. We conclude that somehow the (4+4) subdivision is preferred before a subdivision without subshells. The difference between (8) and (4+4) is the homogeneous Neumann condition acting between subshells, tending to increase the width of the shell and thus increase the energy.

The deeper reason for this preference remains to describe, but the intuition suggests that it relates to the shape or size of the domain occupied by an electron. With subshells electron domains are obtained by subdivision in both radial and azimuthal direction, while without subshells there is only azimuthal/angular subdivision of each shell.

We observe that ionisation energies, which are of similar size in different shells, become increasingly small as compared to ground state energies, and thus are delicate to compute as the difference between the ground state energies of atom and ion.

Recall that a main task for StdQM is to explain the Periodic Table, but the mission does not seem to be complete [2]. It is possible that RealQM can do better.

Figure 10: Atom Simulator Interface

Figure 11: Atom Simulator Output

## 10 From Weak to Strong Reacticity

A basic question of theoretical chemistry is why Gold $Au$ has weak reactivity. To prepare an answer, we start recalling the following electron shell structure for atoms with one valence electron in the outermost shell including $Au$, with E 1st ionization energy according to NIST:

- E = 0.339 Au: 2+8+18+32+18+1
- E = 0.143 Cs: 2+8+18+18+8+1
- E = 0.205 Ag: 2+8+18+18+1
- E = 0.154 R : 2+8+18+8+1
- E = 0.213 Cu: 2+8+18+1
- E = 0.159 K: 2+8+8+1
- E = 0.188 Na: 2+8+1
- E = 0.198 Li: 2+1
- E = 0.500 H: 1

We see that $Cs$ , $Rb$, $K$ and $Na$ all have shell structure ending with 8+1 together with observed small 1st ionization energy and strong reactivity, which can be explained by Coulomb attraction after transfer of the valence electron to form an ionic bond.

We see that $Au$, $Ag$ and $Cu$ all have electron structure ending with 18+1 together with observed large 1st ionization energy and weak reactivity decreasing with number of electrons. Large ionization energy prevents iconic bonds, but the possibility of a covalent bond remains. We now seek a theoretical explanation of the observed fact that $Au$ does not form an $Au_2$ molecule, while $H$ forms $H_2$. What is the essential difference between $Au$ and $H$ both having one valence electron?

According to Standard Quantum Mechanics StdQM as the current canon, the weak reactivity of $Au$ comes from a relativistic effect of inner electrons swirling around the kernel at half the speed of light thus gaining mass and so strongly binding the valence electron into weak reactivity. The same argument would seem to apply to $Cs$ and even more to $Fr$ and so contradict their strong reactivity. To explain the weak reactivity of $Au$ as a relativistic effect does not seem to be real physics.

To seek to understand why $Au$ does not form the molecule $Au_2$, while $H$ forms $H_2$ [14], we let RealQM using the code [24] to compute the energy E1 of an X2 molecule formed by two X atoms with one valence electron outside a shell of radius R to get the following results with E1 minimal at a ceratin distance D1 compared with energy E2 at large distance D2:

- R = 0: E1 =-1.17 at D1 = 1.4, E2 = -1 at D2 = 4.
- R = 0.5: E1 = -0.87 at D1 = 2, E2 = -0.85 at D2 = 4.
- R = 1: E1 = -0.51 at D1 = 2, E2 = - 0.63 at D2 =4.

We make the following connection to H vs Au:

- R = 0: corresponds to H2 with energy release 0.17 under formation of molecule.

- R = 0.5: is intermediate with very small energy release 0.02.
- R = 1: corresponds to Au2 which needs input of 0.12 to form.

By inspecting the figures below it is easy to understand that (i) the covalent bonding of $H_2$ comes from presence of charge density between the kernels, (ii) the non-bonding of $Au_2$ comes from lack of space between the kernels for electron presence.

RealQM also offers an explanation of the large ionization energy of $Au$ (0.339) compared to $Cs$ (0.143) as an effect of the 18+1 shell configuration ending compared to 8+1. Ionization means that the valence electron is dragged away from the atom, which in the shell configuration of RealQM requires to also drag the 18 or 8 inner shell electrons to zero density with an increasing energy cost for 18. The even higher cost for H comes from the fact the the electron of H reaches the kernel, which the valence electrons of $Au$ and $Cs$ do not.

Figure 12: Two $Au$ atoms with large inner shell radius and small binding.

Figure 13: Two $H$ atoms with zero inner shell radius and large binding

Figure 14

## 11 Conclusion

RealQM follows up the early idea of Schrödinger to view electrons as charge densities carrying identity from spatial occupation into a model of atomic physics within classical multi-phase continuum physics, which has linear computational complexity expressed in an essentially 3-line computer code. We show that RealQM captures basic aspects of atomic and molecular physics and so appears to open new possibilities in Molecular Dynamics of large molecules.

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Figure 15

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